

To produce ammonia (NH_3), a stream containing nitrogen (N_2) and hydrogen (H_2) in a 1:1 molar ratio is fed into a reactor. The reactor operates steadily at 100 atm and 700°C . The single reaction taking place in the reactor is: $\text{N}_2 + 3 \text{H}_2 \rightarrow 2 \text{NH}_3$. The reactor converts 23% of the nitrogen entering the reactor into ammonia. There is a single output stream from the reactor. The ammonia production rate is 1200 mol/min of NH_3 .

- (a) Draw a labeled block diagram and calculate the degrees of freedom for the process using p. 105 of the text. Be sure to give values and a brief explanation for 4a, 4b, 5a, 5b, 5c, and 5d.
- (b) Write and solve the appropriate mole balances to find the molar flow rates of all the species exiting the reactor in kmol/hr.
- (c) What are the volume flow rates of the input stream to and the output stream from the reactor in m^3/min ?
- (d) Briefly explain the physical reason for the relative magnitude of your results in part (c).